

This document summarizes the final visualizer delivered as `minimized-bg-final.html`. It's a compact single-file visual tool that renders nested minimized-surface-style backgrounds using SDF primitives, supports HSLA inheritance, compositing modes, hue cycling, and export (PNG / WebM).

Quick facts

- **Single file:** `minimized-bg-final.html` (standalone, no build step). Drop into VS Code and open in browser.
 - **Libraries used (CDN):** Three.js, lil-gui, GSAP.
 - **Shader capacity:** `MAX_TOTAL = 64` (compile-time cap for the GLSL arrays). Adjust and re-save if you need higher capacity.
 - **Runtime logical limits:** `topLevelMax` and `perParentMax` control allowed top-level and per-parent nested minimizers. These are enforced at runtime.
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Key features

1. **Input-agnostic visual heuristics:** uses compact SDF circle/cap primitives in the fragment shader for fast visuals.
 2. **Nested minimizers with color inheritance:** children inherit HSLA from parent unless user supplies an explicit HSLA. Inheritance is transitive.
 3. **Two compositing modes:**
 4. **Back-to-front (default)** — painter's algorithm, predictable layering (children appear on top of parents).
 5. **Front-to-back** — nearest-first compositing with early-out (may be faster for many layers; visually different).
 6. **HSLA passed to shader:** HSLA is normalized in JS and passed to GLSL; hue cycling implemented on GPU via `u_time` and per-minimizer `hueSpeed`.
 7. **Exports:** PNG and WebM (MediaRecorder) via GUI.
 8. **Optional connector:** `window.MinimizerLoader` hook for integrating precomputed SDF/mesh from your minimizer module.
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How to use (short)

1. Open the file in a browser (or serve it locally). GUI appears at top-right.
 2. Add minimizers, select one, click on canvas to place, tweak radius/blend/hue/saturation/lightness/alpha/hueSpeed.
 3. Assign `parentId` to nest a minimizer under a parent (children inherit color if `inherit color` is enabled).
 4. Toggle `Composite` mode to choose compositing ordering. Use `Hue cycling` toggle to enable/disable automatic hue animation.
 5. Export single frame (PNG) or record a short WebM via the `Export WebM` control.
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Notes for developers

- To integrate your algorithmic MinimizerModule, provide a global `window.MinimizerLoader` with `getPreviewMesh()` or `loadPrecomputed(...)` to supply a mesh/SDF for rendering. The visualizer will detect this and can be extended to render the mesh.
 - If you increase `MAX_TOTAL` in the JS, update both shader compile-time value and the typed arrays sizes.
 - The compositor uses premultiplied-alpha semantics to reduce halos.
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Where to go next

- Replace heuristic SDF primitives with precomputed voxel SDFs or meshes from the algorithmic module for physical correctness. Use marching-cubes output loaded into Three.js.
 - Add GPU-based level-set solver (WebGPU/compute) for interactive minimizer computation on the client.
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End of overview.

(If you need a true PDF file, open this document in the canvas and use your browser's "Print to PDF" or VS Code's export.)